

외계행성 대기

Exoplanetary Atmospheres

Lecture 5: Radiative Transfer III — Albedos, Flux Ratios, Phase Curves, and Simple Solutions 2026 October 5 (Monday)

Based on: Chapters 3 and 5 of Seager, *Exoplanet Atmospheres* (2010), and Chapter 2 of Heng,
Exoplanetary Atmospheres: Theoretical Concepts and Foundations (2017)

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Outline – Lecture 5

Planetary Albedos (continued)

Planet-Star Flux Ratios

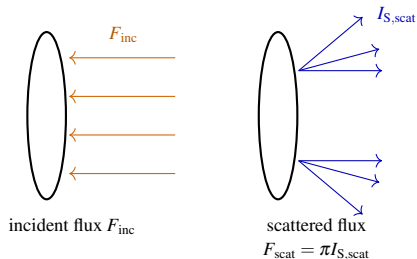
Planetary Phase Curves

Simple Solutions and the Practical Checklist

Clouds and Atmospheric Retrieval

Summary

The Lambert-disk flux and the meaning of I_{inc}



Scattered flux. A Lambert disk scatters with the same intensity $I_{\text{S,scat}}$ into every outgoing direction, so the hemispheric integral of $\cos \theta$ gives $F_{\text{scat}} = \pi I_{\text{S,scat}}$.

Energy conservation. A perfect Lambert disk absorbs nothing:

$$F_{\text{inc}} = F_{\text{scat}} = \pi I_{\text{S,scat}} \implies I_{\text{S,scat}} = \frac{1}{\pi} F_{\text{inc}} \stackrel{\text{define}}{=} I_{\text{inc}}.$$

So I_{inc} is not the intensity of the stellar beam but the intensity the disk scatters back.

Flux at Earth. Seen face-on at distance $D_{\oplus}$,

$$\begin{aligned} F_{\text{L disk}} &= \left(\frac{R_p}{D_{\oplus}} \right)^2 \int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} I_{\text{inc}} \cos \phi \cos^2 \theta \, d\theta \, d\phi \\ &= \left(\frac{R_p}{D_{\oplus}} \right)^2 \pi I_{\text{inc}}, \end{aligned}$$

since $\int \cos \phi \, d\phi = 2$ and $\int \cos^2 \theta \, d\theta = \pi/2$.

A perfect Lambert surface has $p = 1$: energy conservation

Consider one surface element whose normal makes the angle Θ with the direction to the star, and use Seager's description $I_{S,\text{scat}} = I_{\text{inc}}(\theta', \phi') p = p \cos \Theta I_{\text{inc}}(\nu)$.

1. **Flux in.** The stellar flux through the element is foreshortened by $\cos \Theta$; with $I_{\text{inc}} \equiv F_{\text{inc}}/\pi$,

$$F_{\text{in}} = \cos \Theta F_{\text{inc}} = \pi \cos \Theta I_{\text{inc}}(\nu).$$

2. **Flux out.** A Lambert surface scatters with the *same* intensity into every outgoing direction, so the flux leaving it is

$$F_{\text{out}} = \int_0^{2\pi} \int_0^{\pi/2} I_{S,\text{scat}} \cos \theta \sin \theta \, d\theta \, d\phi = \pi I_{S,\text{scat}} = \pi p \cos \Theta I_{\text{inc}}(\nu).$$

3. **Energy conservation.** A *perfect* Lambert surface absorbs nothing, so $F_{\text{out}} = F_{\text{in}}$:

$$\pi p \cos \Theta I_{\text{inc}} = \pi \cos \Theta I_{\text{inc}} \implies \boxed{p = 1}$$

- The two factors of π — the hemispheric integral of $\cos \theta$ and the definition $I_{\text{inc}} \equiv F_{\text{inc}}/\pi$ — cancel, which is why p is simply the reflected fraction: a surface that reflects the fraction A has $p = A$.
- This $p = 1$ is what gave $A_g = 2/3$ for the Lambert sphere, and $A_S = A_g q = 1$: nothing absorbed.

The geometric albedo: general expression

Dividing numerator by denominator,

$$A_g(\nu) = \frac{\int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} I_{S,\text{scat}}(\theta, \phi, \nu) \cos \phi \cos^2 \theta \, d\theta \, d\phi}{\pi I_{\text{inc}}(\nu)}$$

- ▶ Something interesting: the denominator $\pi I_{\text{inc}}(\nu) = F_{\text{inc}}(\nu)$ is the incident stellar flux at the *substellar point*. So A_g may also be described as the ratio of the planet's scattered flux to the incident stellar flux at the substellar point.
- ▶ This is the form to use to compute A_g from the output of a *model atmosphere code*, which delivers the scattered intensity at the planetary surface as a function of location, $I_{S,\text{scat}}(\theta, \phi, \nu)$.
- ▶ The subscript “scat” means that only *scattered* radiation is counted, not radiation that has been absorbed and thermally reradiated; at visible wavelengths, for planets cooler than ~ 1500 K, the distinction is clean.

The geometric albedo: making the physics explicit

For a more conceptual understanding, relate the scattered intensity to the incoming intensity with the dimensionless $p(\Theta)$: the fraction of incident intensity scattered out of the planet into the angle Θ from the incident direction. All the physics of directional and multiple scattering inside the atmosphere is buried in p (it is *not* the single-scattering phase function P of Chapter 5):

$$I_{S,\text{scat}}(\theta, \phi, \nu) = I_{\text{inc}}(\theta', \phi', \nu) p(\Theta),$$

primes for the direction of incidence, no primes after scattering. For **uniform** incident stellar intensity, diluted away from the substellar point by the projection (Fig. 2.3),

$$I_{\text{inc}}(\theta', \phi', \nu) = \hat{\mathbf{n}} \cdot \hat{\mathbf{k}} I_{\text{inc}}(\nu) = \cos \Theta I_{\text{inc}}(\nu) = \cos \theta' \cos \phi' I_{\text{inc}}(\nu),$$

so that $I_{S,\text{scat}} = \cos \theta \cos \phi I_{\text{inc}}(\nu) p(\theta, \phi, \theta', \phi')$. Then $I_{\text{inc}}(\nu)$ cancels between numerator and denominator:

$$A_g(\nu) = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} p(\theta, \phi, \theta', \phi') \cos^2 \phi' \cos^3 \theta' d\theta' d\phi'$$

One factor $\cos \phi' \cos \theta'$ comes from the oblique illumination, the other $\cos \phi' \cos^2 \theta'$ from the flux projection and the solid angle toward the observer.

A_g of a Lambert sphere, and why A_g can exceed 1

For a **Lambert sphere** the intensity is scattered equally in all directions, $p(\Theta) = 1$, and

$$A_g^{\text{Lambert}} = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} \cos^2 \phi \cos^3 \theta \, d\theta \, d\phi = \frac{1}{\pi} \cdot \frac{\pi}{2} \cdot \frac{4}{3} = \boxed{\frac{2}{3}}$$

- ▶ $A_g = 2/3 < 1$ even though a Lambert sphere scatters *all* incident radiation and absorbs none. One third of the light is scattered out of the line of sight: the limb regions are illuminated obliquely *and* seen obliquely, so they contribute less than the corresponding parts of a flat disk.
- ▶ By the same definition, A_g can be greater than 1: it compares the planet not to a [diffusing sphere](#) but to a [perfectly diffusing disk](#). **Enceladus** has $A_g = 1.38$ at visible wavelengths (Verbiscer et al. 2007): fresh ice backscatters strongly, so at $\alpha = 0$ it returns more light toward the observer than a Lambert disk would. Rayleigh-scattering atmospheres also backscatter more than Lambert ($A_g = 3/4$ for a semi-infinite conservative Rayleigh atmosphere).
- ▶ Measurements of A_g have been elusive for exoplanets (upper limits from MOST, CoRoT, Kepler; a few detections such as Kepler-7b with $A_g \simeq 0.3$), but A_g remains the standard way to classify theoretical models.

The fraction scattered toward $\alpha = 0$ is A_g/π .

A_g compares the planet with a Lambert disk; it is *not* the fraction of the intercepted energy sent back toward the star. That fraction per steradian (exercise 3.5) follows in three steps.

1. **Power intercepted.** The planet presents its cross section to the star: $P_{\text{inc}} = \pi R_p^2 F_{\text{inc}}$.
2. **Power sent toward the observer.** By definition of A_g , with $\pi I_{\text{inc}} = F_{\text{inc}}$, the flux received at distance $D_{\oplus}$ at $\alpha = 0$ is $F_{\oplus} = A_g (R_p/D_{\oplus})^2 F_{\text{inc}}$. Multiplying by $D_{\oplus}^2$ turns this into power per unit solid angle leaving the planet in that direction: $(dP/d\Omega)_{\alpha=0} = D_{\oplus}^2 F_{\oplus} = A_g R_p^2 F_{\text{inc}}$.
3. **The ratio.** Dividing by the intercepted power,

$$\frac{1}{P_{\text{inc}}} \left(\frac{dP}{d\Omega} \right)_{\alpha=0} = \frac{A_g R_p^2 F_{\text{inc}}}{\pi R_p^2 F_{\text{inc}}} = \frac{A_g}{\pi} \text{ sr}^{-1}.$$

- It is a fraction *per steradian*. Energy conservation limits only its integral over all directions ($\leq A_B \leq 1$), which is why a strong backscatterer can have $A_g > 1$.
- Checks: a Lambert disk facing the star ($A_g = 1$) sends $1/\pi$ per sr along its normal, as the cosine law requires; a planet scattering everything isotropically sends $1/(4\pi)$ per sr, i.e. $A_g = 1/4$; a Lambert sphere ($A_g = 2/3$) sends $2/(3\pi)$, which is $8/3$ times the isotropic value — the same $8/3$ as the dayside factor f , from the same double-cos ϑ integral. Enceladus: $1.38/\pi \simeq 0.44$ per sr.

The Bond albedo A_B and the spherical albedo $A_S(\nu)$

- ▶ The **bond albedo** A_B is the fraction of incident stellar *energy* scattered back into space, including radiation into *all* directions and at *all* frequencies. By definition $A_B \leq 1$.
- ▶ The **spherical albedo** $A_S(\nu)$ is the same quantity at a specific frequency. The Bond albedo is the spherical albedo averaged over frequency *weighted by the incident stellar spectrum*:

$$A_B = \frac{\int_0^\infty A_S(\nu) F_{S,*}(\nu) d\nu}{\int_0^\infty F_{S,*}(\nu) d\nu}$$

(Seager writes $A_B = \int A_S d\nu$ as shorthand and states that the weighting by the incident radiation is implied; Heng's Eq. 2.10 gives the weighted form explicitly.)

- ▶ Consequently A_B depends on the *host star's spectrum*: the same planet has a different Bond albedo around a G star and around an M star, because an M star emits most of its light in the near-IR where the atmosphere may be dark.

To derive $A_S(\nu)$ we must account for the light scattered at *every* phase angle, not just at $\alpha = 0$. At each α only a portion of the hemisphere facing the observer is illuminated (next slide).

Scattered flux at phase angle α

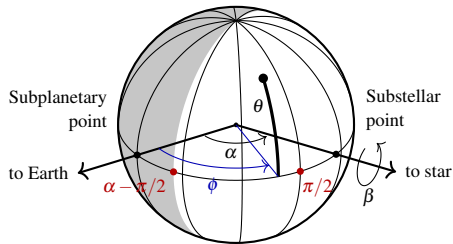


Figure 1: Coordinates for the phase-dependent scattered flux. θ is latitude from the intensity equator, ϕ longitude from the sub-Earth point, α the phase angle, β the azimuth of the observer about the star direction (next slide). After Sobolev (1975); redrawn after Seager (2010), Fig. 3.6.

Measure longitude ϕ from the sub-Earth (subplanetary) point. The substellar point is then at longitude α , and a surface element at (θ, ϕ) is illuminated with the projection $\cos(\alpha - \phi) \cos \theta$ and seen with the projection $\cos \phi \cos \theta$. Only the longitudes $\alpha - \pi/2 < \phi < \pi/2$ are both lit and visible. The scattered flux toward the observer is

$$F_{S,\alpha}(\nu) = \int_{\alpha-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} I_{\text{inc}}(\theta', \phi') p(\theta, \phi, \theta', \phi') \times \cos(\alpha - \phi') \cos \phi' \cos^3 \theta' d\theta' d\phi',$$

and for uniform incident intensity we write

$$F_{S,\alpha}(\nu) = I_{\text{inc}}(\nu) \Psi_{\alpha}(\nu),$$

where Ψ_{α} is the fraction of scattered radiation in the direction of the observer as a function of α . At $\alpha = 0$ it reduces to the geometric-albedo integral: $\Psi_{\alpha=0} = \pi A_g$.

The spherical albedo: integrating over all directions

To count the energy scattered into all directions, surround the planet with a sphere and use α as the polar angle and β as the azimuth (azimuthal symmetry assumed). Both are measured at the planet's center: α is the angle between the direction to the star and the direction to a point on the sphere — the phase angle of an observer there — and β the azimuth about the star direction. Each surface element reflects only into its own outward hemisphere, but the curved lit face as a whole sends light toward every α from 0 to π (with $\Psi_\alpha \rightarrow 0$ as $\alpha \rightarrow \pi$). The total scattered energy per unit time per frequency is

$$E_{\text{scat}}(\nu) = I_{\text{inc}}(\nu) \int_0^{2\pi} \int_0^\pi \Psi_\alpha(\nu) R_p^2 \sin \alpha \, d\alpha \, d\beta = 2\pi R_p^2 I_{\text{inc}}(\nu) \int_0^\pi \Psi_\alpha(\nu) \sin \alpha \, d\alpha.$$

The incident energy (Chapter 2) is $E_{\text{inc}}(\nu) = \pi R_p^2 \cdot \pi I_{\text{inc}}(\nu)$. Their ratio is the spherical albedo:

$$A_S(\nu) = \frac{E_{\text{scat}}(\nu)}{E_{\text{inc}}(\nu)} = \frac{2}{\pi} \int_0^\pi \Psi_\alpha(\nu) \sin \alpha \, d\alpha$$

- ▶ A_S needs Ψ_α at *all* phase angles. From Earth we cannot see all phases of most planets, so A_S (and A_B) are not directly measurable: we need a way to estimate them from the one thing we can hope to measure near full phase, A_g .
- ▶ Note that, in the factor $2/\pi$, the factor of 2 comes from the sphere-to-disk conversion, while the factor of π arises from the flux.

Relating A_S to A_g : the phase integral q

Using $\Psi_{\alpha=0}(v) = \pi A_g(v)$ (next slide), rewrite the spherical albedo as

$$A_S(v) = \frac{\pi A_g(v)}{\Psi_{\alpha=0}(v)} \cdot \frac{2}{\pi} \int_0^\pi \Psi_\alpha(v) \sin \alpha \, d\alpha \quad \Longrightarrow \quad \boxed{A_S(v) = A_g(v) q(v)}$$

with the **phase integral**

$$q(v) \equiv \frac{A_S(v)}{A_g(v)} = 2 \int_0^\pi \Phi_\alpha(v) \sin \alpha \, d\alpha, \quad \Phi_\alpha(v) \equiv \frac{\Psi_\alpha(v)}{\Psi_{\alpha=0}(v)}$$

where Φ_α is the **phase function**: the scattered flux versus phase angle normalized to full phase (this is the phase curve of the next lecture).

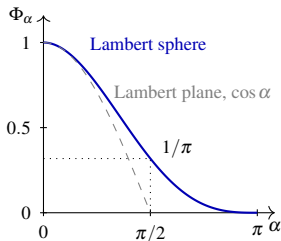
- ▶ A_S can be *estimated* from a measured A_g and a theoretical q . Heng writes the same thing as $A_S = A_g P_g$.
- ▶ **Lambert sphere:** $\Phi_\alpha = \frac{1}{\pi} [\sin \alpha + (\pi - \alpha) \cos \alpha]$ (derived on the next slide), so $q = 3/2$ and

$$A_g = \frac{2}{3} A_S, \quad A_S = A_B = 1, \quad A_g = \frac{2}{3}.$$

All light is scattered back to space, so the spherical albedo is 1; the geometric albedo counts only the light scattered toward the observer at $\alpha = 0$ and is less.

- ▶ **Rayleigh scattering:** $q = 4/3$; a Lambert *plane* (exercise 3.6): $\Phi_\alpha = \cos \alpha$. Real planets: $q \simeq 1.2$ – 1.7 .

The Lambert sphere: deriving Φ_α and q



Phase function of a Lambert sphere. At quadrature it is only $1/\pi$ as bright as at full phase.

$$\begin{aligned} \Phi_0 &= 1, \Phi_{\pi/2} = 1/\pi, \Phi_\pi = 0. \text{ With} \\ \int_0^\pi \sin^2 \alpha \, d\alpha &= \pi/2 \text{ and} \\ \int_0^\pi (\pi - \alpha) \sin \alpha \cos \alpha \, d\alpha &= \pi/4: \\ q &= 2 \int_0^\pi \Phi_\alpha \sin \alpha \, d\alpha = \frac{2}{\pi} \left(\frac{\pi}{2} + \frac{\pi}{4} \right) = \frac{3}{2}, \\ A_S &= A_g q = \frac{2}{3} \cdot \frac{3}{2} = 1. \end{aligned}$$

In general $\Psi_{\alpha=0} = \pi A_g$. Seager's Eq. (3.30) prints A_g/π , which would make her Eq. (3.31) inconsistent with Eq. (3.29).

For a perfect Lambert surface $p = 1$, and the integral for Ψ_α separates into a latitude and a longitude part:

$$\Psi_\alpha = \int_{-\pi/2}^{\pi/2} \cos^3 \theta \, d\theta \int_{\alpha-\pi/2}^{\pi/2} \cos(\alpha - \phi) \cos \phi \, d\phi.$$

- **Latitude:** $\int_{-\pi/2}^{\pi/2} \cos^3 \theta \, d\theta = \frac{4}{3}$.
- **Longitude:** with $\cos(\alpha - \phi) \cos \phi = \frac{1}{2} [\cos \alpha + \cos(\alpha - 2\phi)]$ over the lit and visible range of width $\pi - \alpha$,

$$\begin{aligned} \int_{\alpha-\pi/2}^{\pi/2} \cos(\alpha - \phi) \cos \phi \, d\phi &= \frac{1}{2} \left[(\pi - \alpha) \cos \alpha + \left(-\frac{1}{2} \sin(\alpha - 2\phi) \right) \Big|_{\alpha-\pi/2}^{\pi/2} \right] \\ &= \frac{1}{2} [\sin \alpha + (\pi - \alpha) \cos \alpha]. \end{aligned}$$

Hence

$$\Psi_\alpha = \frac{2}{3} [\sin \alpha + (\pi - \alpha) \cos \alpha], \quad \Psi_{\alpha=0} = \frac{2\pi}{3} = \pi A_g \Rightarrow A_g = \frac{2}{3},$$

$$\Phi_\alpha = \frac{1}{\pi} [\sin \alpha + (\pi - \alpha) \cos \alpha]$$

Bond versus geometric albedo for the giant planets

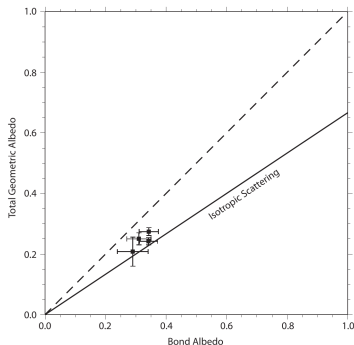


Figure 2: Total (wavelength-integrated) geometric albedo versus Bond albedo for a Lambert sphere (solid, $A_g/A_B = 2/3$) and for Uranus, Neptune, Saturn, Jupiter (points, in order of increasing A_B). The dashed line is $A_g = A_B$. Adapted from Rowe et al. (2006); reproduced from Seager (2010), Fig. 3.7.

- ▶ All four giants lie *above* the Lambert (isotropic scattering) line: their atmospheres backscatter more than a Lambert sphere, so $A_g/A_B > 2/3$ ($q < 3/2$).
- ▶ The wedge between the Lambert line and the line $A_g = A_B$ bounds the photometric properties of most spherical bodies with deep atmospheres (Rayleigh scattering, clouds, dust). **A planet with deep atmosphere must lie to the right of the dashed line.**
- ▶ Practical use: a measured (or upper-limit) A_g of an exoplanet, e.g. the MOST limit $A_g < 0.25$ for HD 209458b, can be converted to a range of A_B , hence to a range of T_{eq} .

The apparent albedo: the one an exoplanet can give us

The geometric, spherical, and Bond albedos are unsatisfactory for exoplanets because *none can be measured*:

- ▶ A_g is defined at full phase ($\alpha = 0$), when the planet is behind the star and unobservable (for transiting planets the eclipse depth measured just before/after eclipse is the practical substitute, at a small α).
- ▶ A_S and A_B require all phase angles. With direct imaging, phases where the planet projects too close to the star are inaccessible, and the planet does not go through all phases at all unless the orbit is edge-on.

Apparent albedo. The ratio of the scattered flux emerging from the planet toward the observer to the scattered flux from a *perfectly reflecting Lambert sphere* of the same size, at the same phase angle, at the same distance:

$$A_{\text{app}}(\alpha, \nu) = \frac{F_{S,\alpha}(\nu)}{F_{S,\alpha}^{\text{Lambert}}(\nu)} = \frac{A_g(\nu) \Phi_\alpha(\nu)}{\frac{2}{3} \Phi_\alpha^{\text{Lambert}}} \quad (\text{exercise 3.9}).$$

- ▶ It is the planet's reflectance in the direction of the observer at a given time, a function of phase angle and frequency.
- ▶ It is the only albedo that can be derived from a *single* observation of a planet of known size — a directly imaged planet at known separation and phase.

The emergent flux

The emergent surface flux follows from the flux definition of Chapter 2 applied to the outgoing hemisphere, $F = 2\pi \int_0^1 \mu I(0, \mu) d\mu$ (no incoming radiation at the top in the thermal infrared):

$$F(0, \nu) = 2\pi \int_0^1 \int_0^\infty S(\tau_\nu, \mu, \nu) e^{-\tau_\nu/\mu} d\tau_\nu d\mu$$

- ▶ If the source function $S(\tau, \mu, \nu)$ is *known*, the emergent intensity and the emergent flux — the planet's spectrum — can be computed directly from these two integrals. The μ integral of $e^{-\tau/\mu}$ produces the exponential integral $E_2(\tau)$, so $F(0, \nu) = 2\pi \int_0^\infty S(\tau_\nu) E_2(\tau_\nu) d\tau_\nu$ for an isotropic S .
- ▶ For $S = B(T(\tau))$ (pure absorption, LTE) this is the standard *forward model* of an emission spectrum: choose $T(\tau_\nu)$, i.e. a temperature–pressure profile and an opacity function, and integrate.
- ▶ In many cases, however, this straightforward solution is not possible. The main complication is that the source function itself depends on $I(\tau, \mu, \nu)$, the quantity we are trying to solve for (next slide).

Planet-Star Flux Ratios

An estimate of the planet-to-star flux ratio tells us whether a given planet and star can be detected with a given telescope, at which wavelengths, and with what precision. Previous slides gave us the ingredients: surface fluxes, the flux at Earth, temperatures, and albedos. We now assemble them.

- ▶ Assume star and planet radiate as blackbodies (a first estimate; brightness temperatures refine it).
- ▶ A planet has **two spectral peaks**: a short-wavelength peak of *scattered starlight*, at the same wavelength as the star, and a long-wavelength peak of *thermal emission* at the planet's own temperature.
- ▶ During transit a third quantity appears: the *transmission* flux ratio, the fraction of starlight blocked by the atmospheric annulus.

The planet flux at Earth is $F_{\oplus,p} = F_{S,p}(R_p/D_{\oplus})^2$ and the stellar flux at Earth is $F_{\oplus,*} = F_{S,*}(R_*/D_{\oplus})^2$, so the distance cancels:

$$\boxed{\frac{F_{\oplus,p}}{F_{\oplus,*}} = \frac{F_{S,p} R_p^2}{F_{S,*} R_*^2}}$$

valid for total fluxes and for frequency-dependent fluxes alike.

Spectra of the Solar System seen from 33 light years

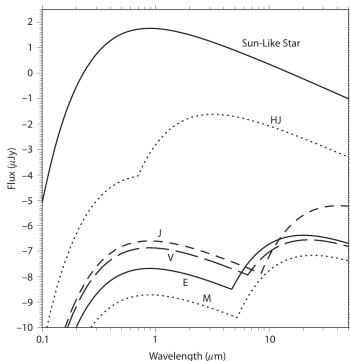


Figure 3: Approximate blackbody spectra (in $10^{-26} \text{ W m}^{-2} \text{ Hz}^{-1}$) of the Sun (5750 K), Jupiter, Venus, Earth, Mars, and a representative hot Jupiter, seen from 33 light years (10 pc). Reproduced from Seager (2010), Fig. 3.8.

- ▶ Each planet shows the two peaks: reflected light near $0.5 \mu\text{m}$ (shape of the stellar spectrum, scaled by $A_g R_p^2 / a^2$) and thermal emission peaking at $\lambda_{\text{max}} \simeq 2898 \mu\text{m K} / T$: $\sim 10 \mu\text{m}$ for Earth and Venus, $\sim 25 \mu\text{m}$ for Jupiter, $\sim 2 \mu\text{m}$ for a representative hot Jupiter.
- ▶ The planet-star contrast is large at *all* wavelengths, but the thermal peak of a hot Jupiter reaches within 10^{-3} of the star, the Earth's thermal peak within 10^{-7} , while in the visible the contrast is 10^{-9} – 10^{-10} .
- ▶ Venus is brighter than Earth in reflected light (high albedo, closer to the Sun) but similar in the thermal infrared: its cloud tops radiate at $T_{\text{eff}} \simeq 230 \text{ K}$.
- ▶ This high contrast is *the* primary challenge for detecting radiation from exoplanets.

Thermal emission flux ratio

On the illuminated side, equate the planet's thermal emission with the absorbed stellar radiation — the energy balance with the redistribution factor f , $4\pi R_p^2 F_{S,p}/f = (1 - A_B) F_{S,*} (R_*/a)^2 \pi R_p^2$ — to get the *total* flux ratio

$$\frac{F_{\oplus,p}}{F_{\oplus,*}} = \frac{F_{S,p} R_p^2}{F_{S,*} R_*^2} = \left(\frac{R_p}{a} \right)^2 \frac{f}{4} (1 - A_B)$$

- ▶ Here f is the correction factor to 4π that describes the unknown redistribution of the absorbed stellar radiation, and the fluxes are total fluxes, integrated over all frequencies.
- ▶ The ratio depends on R_p/a , not on R_p/R_* : the planet reradiates the fraction $\pi R_p^2/4\pi a^2$ of the stellar luminosity it intercepts, times $(1 - A_B)$, times the redistribution factor.
- ▶ For a hot Jupiter at $a = 0.05$ AU with $A_B = 0.1$ and $f = 1$: $(R_J/a)^2 \times 0.9/4 \simeq 2 \times 10^{-5}$ bolometrically. Yet the observed ratio in an infrared band is $\sim 10^{-3}$, because the planet's emission is concentrated at long wavelengths where the star is faint — which is why we need a frequency-dependent estimate.

Thermal emission flux ratio at a given frequency

A more useful planet-star flux ratio estimate is **frequency dependent**, because telescope observations are limited to a fixed frequency or wavelength range. Approximate the star and planet surface fluxes by blackbody fluxes $\pi B(T, \nu)$, with T the effective temperature of the star and the equilibrium temperature of the planet:

$$\frac{F_{\oplus,p}(\nu)}{F_{\oplus,*}(\nu)} = \frac{F_{S,p}(\nu)R_p^2}{F_{S,*}(\nu)R_*^2} = \frac{e^{h\nu/k_B T_{\text{eff},*}} - 1}{e^{h\nu/k_B T_{\text{eq},p}} - 1} \frac{R_p^2}{R_*^2}$$

- ▶ The ν^3 prefactors of the two Planck functions cancel; only the exponential factors and the area ratio remain.
- ▶ In the Wien tail of the planet ($h\nu \gg k_B T_{\text{eq}}$) the ratio is $\simeq e^{-h\nu(1/k_B T_{\text{eq}} - 1/k_B T_{\text{eff},*})} (R_p/R_*)^2$, exponentially small; it rises steeply toward the infrared and saturates in the Rayleigh–Jeans regime (next slide).
- ▶ Note that it is now R_p/R_* that matters, not R_p/a : at a fixed planet temperature, a smaller star gives a larger contrast. This is the same “M-dwarf opportunity” as for transits.
- ▶ Example: HD 189733b at $8 \mu\text{m}$ with $R_p/R_* = 0.155$, $T_{\text{eq}} = 1200 \text{ K}$, $T_{\text{eff},*} = 5000 \text{ K}$ gives $(e^{3.6} - 1)/(e^{1.5} - 1) \times 0.024 = 0.0034$, close to the observed eclipse depth of 0.34% (Lecture 1).

The Rayleigh–Jeans limit

At mid-infrared wavelengths a further simplification applies: for reasonable planet and star temperatures both blackbody fluxes are in the **Rayleigh–Jeans** region, $h\nu/k_B T \ll 1$. With $e^x - 1 \simeq x$ the Planck function becomes

$$\pi B(T, \nu) \simeq \pi \frac{2\nu^2}{c^2} k_B T,$$

linear in T , and the flux ratio becomes

$$\boxed{\frac{F_{\oplus,p}(\nu)}{F_{\oplus,*}(\nu)} = \frac{F_{S,p}(\nu) R_p^2}{F_{S,*}(\nu) R_*^2} \simeq \frac{T_{\text{eq},p}}{T_{\text{eff},*}} \left(\frac{R_p}{R_*} \right)^2}$$

in the frequency range where the approximation is valid.

- ▶ This is the *maximum* contrast the thermal emission can reach: the temperature ratio times the area ratio.
- ▶ **Hot Jupiter:** $(1500/5800) \times (0.1)^2 \simeq 2.6 \times 10^{-3}$, approached already at $\lambda \gtrsim 10 \mu\text{m}$ (for $T_p = 1500 \text{ K}$, $h\nu/k_B T = 1$ at $\lambda = 9.6 \mu\text{m}$).
- ▶ **Earth:** $(255/5800) \times (0.0092)^2 \simeq 3.7 \times 10^{-6}$, reached only at $\lambda \gtrsim 100 \mu\text{m}$; at $10 \mu\text{m}$ the actual ratio is $\sim 10^{-7}$ because Earth is still on the Wien side.
- ▶ The Rayleigh–Jeans form makes explicit why the infrared is the place to detect thermal emission: the contrast saturates at T_p/T_* (independent of wavelength) instead of decaying exponentially.

Using brightness temperatures instead of blackbodies

Real planets and stars deviate from blackbodies, sometimes strongly (the M-dwarf spectrum of last lecture, molecular bands in the planet). In the Rayleigh–Jeans regime, the flux ratio can be expressed in terms of the **brightness temperatures** at each frequency:

$$\boxed{\frac{F_{\oplus,p}(\nu)}{F_{\oplus,*}(\nu)} = \frac{F_{S,p}(\nu)R_p^2}{F_{S,*}(\nu)R_*^2} = \frac{T_{b,p}(\nu)}{T_{b,*}(\nu)} \left(\frac{R_p}{R_*}\right)^2} \quad (\text{Rayleigh–Jeans regime}),$$

and, outside the Rayleigh–Jeans regime, the ratio of Planck functions evaluated at $T_{b,p}(\nu)$ and $T_{b,*}(\nu)$.

- ▶ This is exactly how a measured **secondary-eclipse depth** is turned into a dayside brightness temperature: with R_p/R_* from the transit and a stellar model for $T_{b,*}(\nu)$, invert the Planck function for $T_{b,p}(\nu)$.
- ▶ The stellar brightness temperature matters: at $3.6\text{--}8\text{ }\mu\text{m}$ a K or M dwarf has $T_{b,*}$ well below its T_{eff} because of molecular absorption, which raises the apparent planet-to-star contrast.
- ▶ $T_{b,p}(\nu)$ across several bands maps out the emission spectrum: bands with lower T_b are absorption features formed high in a (normally stratified) atmosphere; bands with higher T_b indicate a temperature inversion.

Scattered-light flux ratio at full phase ($\alpha = 0$)

Scattered starlight is typically at visible wavelengths (extending to longer wavelengths for colder planets). At $\alpha = 0$, start from the thermal flux ratio and replace the absorbed-and-reradiated factor by a scattered one:

$$\frac{f}{4}(1 - A_B) \longrightarrow \frac{g}{4}A_S(\nu),$$

where $A_S(\nu)$ is the spherical albedo (we want a wavelength-dependent ratio; monochromatic scattering assumed) and g , like f , is a correction to 4π : the solid angle into which the incident radiation is scattered compared to 4π . Scattered light cannot be advected around the planet; it leaves the illuminated hemisphere, with a directional pattern. Thus

$$\frac{F_{\oplus,p}}{F_{\oplus,*}} = \left(\frac{R_p}{a}\right)^2 \frac{f}{4}(1 - A_B) \quad \Rightarrow \quad \frac{F_{\oplus,p}(\nu)}{F_{\oplus,*}(\nu)} = \left(\frac{R_p}{a}\right)^2 \frac{g}{4}A_S(\nu).$$

The critical step is g : the ratio of energy emitted at zero phase angle to that emitted at all phase angles. With E_{scat} from the spherical-albedo derivation and Ψ_α the fraction scattered toward the observer,

$$\frac{g}{4\pi} = \frac{E_{\text{scat},\alpha=0}}{E_{\text{scat}}} = \frac{\Psi_{\alpha=0}(\nu)}{2\pi \int_0^\pi \Psi_\alpha(\nu) \sin \alpha d\alpha} = \frac{1}{\pi q(\nu)} = \frac{A_g(\nu)}{\pi A_S(\nu)} \quad \Rightarrow \quad \boxed{g = \frac{4A_g(\nu)}{A_S(\nu)}}$$

using $\Psi_{\alpha=0} = \pi A_g$, the phase function Φ_α , and the phase integral $q = A_S/A_g$.

Scattered-light flux ratio: the result and numbers

Substituting g , the spherical albedo cancels and the full-phase visible flux ratio is

$$\frac{F_{\oplus,p}(\nu)}{F_{\oplus,*}(\nu)} = A_g(\nu) \frac{R_p^2}{a^2}$$

- Note that the visible flux ratio at full phase involves the *geometric* albedo, which describes the light scattered toward the observer at $\alpha = 0$, not the spherical or Bond albedo, which describe the energy scattered in all directions. That is precisely what g converts.
- The ratio falls as a^{-2} and does not involve R_* at all (for a fixed stellar flux at the planet, a bigger star only means a bigger a for the same temperature).
- **Jupiter:** $A_g(500\text{ nm}) = 0.5$, $a = 5.2\text{ AU}$, $R_p = R_J$ gives $F_{\oplus,p}/F_{\oplus,*} = 4.4 \times 10^{-9}$.
- **Scaled to 0.05 AU:** 10^4 times higher, 4.4×10^{-4} . However, the actual hot Jupiters are darker: Jupiter is bright because of its icy NH_3 and H_2O clouds, whereas hot Jupiters lack bright clouds and have strong Na/K absorption, so $A_g \lesssim 0.1$ and the ratio is $\lesssim 10^{-4}$ — consistent with the Kepler and MOST upper limits and detections.
- **Earth:** $A_g(500\text{--}800\text{ nm}) = 0.2$, $a = 1\text{ AU}$: 2.1×10^{-10} at $\alpha = 0$ (the number quoted in Lecture 1).

Transmission flux ratio

During transit, starlight passes through the planet atmosphere and picks up its spectral features: the **transmission spectrum**. What is measured is the total (star + planet) flux in transit divided by the out-of-transit flux. Let R_p be the radius where the planet is opaque at all wavelengths of interest and A_H the additional radial extent of the atmosphere. Splitting the transmission spectrum ($F_{\text{intrans},\oplus}$) into the unobscured stellar disk (I_*), the opaque planet ($I_p = 0$), and the atmospheric annulus (I_{atm}):

$$F_{\text{intrans},\oplus}(\nu) = \int_0^{2\pi} \int_{(R_p+A_H)/D_\oplus}^{R_*/D_\oplus} I_* \cos \omega \sin \omega d\omega d\phi + \int_0^{2\pi} \int_{R_p/D_\oplus}^{(R_p+A_H)/D_\oplus} I_{\text{atm}} \cos \omega \sin \omega d\omega d\phi.$$

Dividing by the out-of-transit stellar flux $F_{*,\oplus} = F_{S,*}(R_*/D_\oplus)^2$ and assuming $A_H \ll R_p$,

$$\frac{F_{\text{intrans},\oplus}(\nu)}{F_{*,\oplus}(\nu)} = 1 - \left(\frac{R_p}{R_*}\right)^2 - \frac{2R_p A_H}{R_*^2} \left[1 - \frac{F_{S,p,\text{trans}}(\nu)}{F_{S,*,\text{trans}}(\nu)}\right]$$

- (1) First term: the star, (2) second: the opaque planetary disk (the **transit depth**), (3) third: the annulus $2\pi R_p A_H$ over the stellar disk πR_*^2 , times the fraction of starlight the annulus *blocks*.

$F_{S,p,\text{trans}}/F_{S,*,\text{trans}} = 1 - e^{-\tau}$ is the transmitted fraction, computed from the slab geometry.

How large is the transmission signal?

The maximum signal occurs where the annulus is completely opaque ($F_{S,p,\text{trans}} = 0$):

$$\frac{F_{p,\oplus,\text{trans}}(v)}{F_{*,\oplus}(v)} = \frac{2R_p A_H}{R_*^2} \left[1 - \frac{F_{S,p,\text{trans}}}{F_{S,*,\text{trans}}} \right] \approx \frac{2R_p A_H}{R_*^2}$$

- ▶ The choice of A_H does not matter as long as it is large enough: above some height there is no atmosphere left to attenuate, the transmitted fraction is 1, and the bracket vanishes. What matters is the height at which the slant path becomes opaque, a few scale heights above the opaque radius.
- ▶ **Hot Jupiter** around a Sun-sized star: an H/He atmosphere with $T_{\text{eq}} \simeq 1000$ K has a scale height $H \simeq 500$ km (Chapter 9). Taking the atmosphere to extend $A_H = 5H$:
 $2R_p A_H / R_*^2 = 2 \times (7.1 \times 10^4) \times (2.5 \times 10^3) / (7 \times 10^5)^2 \simeq 7 \times 10^{-4}$, i.e. of order 10^{-3} (Seager quotes 1.5×10^{-3}).
- ▶ One part per thousand may seem very small. In fact it is a *large* number in the field of exoplanet atmosphere detection: compare with the $\lesssim 10^{-4}$ reflected-light and $\sim 10^{-3}$ thermal ratios above, and note that it requires no separation of the planet's light from the star's, only precise photometry in and out of transit.

Transmission signals of cooler planets, and the three channels

- ▶ **Jupiter analog** (a Jupiter-sized planet in a Jupiter-like orbit, $T_{\text{eq}} \simeq 120 \text{ K}$, $H = 24 \text{ km}$): the estimated transmission strength is only $\sim 4\text{--}7 \times 10^{-5}$. Cold planets have small scale heights and thus weak transmission spectra, even though their transit depth is the same.
- ▶ **Earth twin** ($H = 8 \text{ km}$, N_2 atmosphere): $\sim 2 \times 10^{-7}$ (Lecture 1). **This is why terrestrial-planet transmission spectroscopy targets M dwarfs, where R_*^2 is 25–100 times smaller.**
- ▶ The transmission technique has been very successful for hot Jupiters around bright stars: Na, K, H_2O , CH_4 , CO, CO_2 , He, and escaping H have all been identified this way, and JWST now routinely delivers transmission spectra of sub-Neptunes.

The three observational channels and what controls each:

channel	flux ratio	key planet property	geometric factor
thermal emission	$(T_{\text{b},p}/T_{\text{b},*})(R_p/R_*)^2$	temperature $T_{\text{b}}(\nu)$	$(R_p/R_*)^2$
reflected light	$A_g R_p^2/a^2$	geometric albedo $A_g(\nu)$	$(R_p/a)^2$
transmission	$2R_p A_H/R_*^2$	scale height H	R_p/R_*^2

Each channel favors different planets: emission and transmission favor hot, large planets around small stars; reflection favors close-in planets but is limited by their low albedos.

Visible-wavelength phase curves

An exoplanet goes through illumination phases like the Moon. We want the flux ratio not only at $\alpha = 0$ but as a function of α . Recall the normalized phase function

$$\Phi_{\alpha}(v) = \frac{\Psi_{\alpha}(v)}{\Psi_{\alpha=0}(v)}, \quad \Psi_{\alpha}(v) = \int_{\alpha-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} I_{\text{inc}}(\theta', \phi', v) p(\theta, \phi, \theta', \phi') \cos(\alpha - \phi') \cos \phi' \cos^3 \theta' d\theta' d\phi'.$$

The Lambert sphere is the one case with an analytic phase curve. With $p = 1$ and uniform incident intensity, the θ integral gives $4/3$ and the ϕ integral gives $\int_{\alpha-\pi/2}^{\pi/2} \cos(\alpha - \phi) \cos \phi d\phi = \frac{1}{2} [\sin \alpha + (\pi - \alpha) \cos \alpha]$, so after normalizing to $\alpha = 0$ (where the integral is $\pi/2$),

$$\Phi_{\alpha} = \frac{1}{\pi} [\sin \alpha + (\pi - \alpha) \cos \alpha]$$

frequency independent, with $\Phi_0 = 1$, $\Phi_{\pi/2} = 1/\pi$, $\Phi_{\pi} = 0$.

- ▶ At quadrature ($\alpha = 90^\circ$) a Lambert sphere is $1/\pi \simeq 0.32$ as bright as at full phase, *less than half* (exercise 3.7): the visible illuminated half-disk is lit obliquely near the terminator and its brightest region (the substellar point) sits at the limb, seen at grazing angle.
- ▶ The phase integral follows: $q = 2 \int_0^{\pi} \Phi_{\alpha} \sin \alpha d\alpha = 3/2$.

Phase curves of Solar System bodies

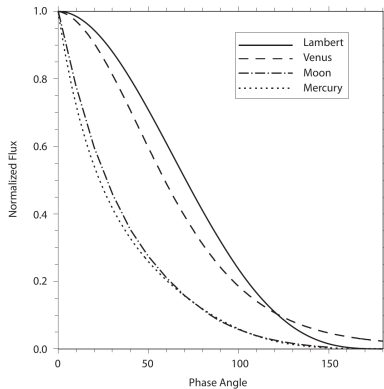


Figure 4: Visible-wavelength phase curves of Venus, Mercury, and the Moon, with the Lambert sphere (rightmost solid line). From formulae in Cox (2000); reproduced from Seager (2010), Fig. 3.9.

- ▶ **Venus**, with its thick cloud deck, follows the Lambert curve closely out to $\alpha \sim 100^\circ$ and then falls faster (forward scattering by cloud droplets brightens it again at very large α , not shown).
- ▶ The **Moon** and **Mercury** — airless, rough, regolith-covered — fall off much faster: their surfaces backscatter (the *opposition surge* near $\alpha = 0$) and shadowing in the rough surface darkens the oblique views. At quadrature the Moon is only ~ 0.1 of its full-phase brightness, not 0.32.
- ▶ Lesson: the phase curve shape encodes the *scattering physics* (clouds versus bare surface, particle size, surface roughness). For exoplanets it is a second observable beyond A_g .

Flux ratio at all phases and an Earth example

The normalized phase curve combined with the full-phase flux ratio gives the planet-star flux ratio at any phase:

$$\frac{F_{p,\oplus,\alpha}(\nu)}{F_{*,\oplus}(\nu)} = A_g(\nu) \frac{R_p^2}{a^2} \Phi_\alpha(\nu)$$

Earth orbiting the Sun as an exoplanet.

- ▶ At $\alpha = 0$ with $A_g(500\text{--}800\text{nm}) = 0.2$, $a = 1\text{ AU}$, $R_p = R_\oplus$: 2.1×10^{-10} . Earth is almost ten billion times fainter than the Sun — and $\alpha = 0$ is when the planet is directly behind the star.
- ▶ At **maximum elongation** ($\alpha = 90^\circ$), the natural configuration for direct imaging, approximating Earth as a Lambert sphere: $\Phi_{90^\circ} = 1/\pi$ and the ratio is 1.2×10^{-10} , a factor π lower. (Earth's real phase curve is not Lambertian: ocean glint and forward-scattering clouds modify it.)
- ▶ For an edge-on orbit the full range $0 < \alpha < 180^\circ$ is swept once per orbit; the amplitude of the reflected-light phase curve is $A_g R_p^2 / a^2$ and its *shape* constrains Φ_α , hence q and A_g . For inclination i , the accessible range is $90^\circ - i \leq \alpha \leq 90^\circ + i$ (exercise 3.8).

Infrared phase curves

Phase variation also occurs at *infrared* wavelengths, from thermal emission, and not just at visible wavelengths from scattered starlight. Thermal phase variation arises when the emission is dominated by reemitted absorbed starlight rather than by internal energy.

- ▶ The prime examples are the **tidally locked** (synchronously rotating) hot Jupiters, hot Neptunes, and hot Earths with $a < 0.05$ AU, strongly heated by their stars. They present the same face to the star at all times, just as the Moon does to Earth.
- ▶ The permanent day and night sides set up a temperature gradient that atmospheric circulation may or may not even out. How the absorbed stellar energy is circulated is a complex topic (Chapter 10).

Simplest case: a hot tidally locked planet where the incident stellar energy is absorbed in one layer, instantaneously heats that surface element, and is reemitted, with no advection. Thermal emission is isotropic, so each element reradiates like a Lambert scatterer and the problem is solved in analogy with the Lambert sphere: the normalized thermal phase curve is again

$$\Phi_{\alpha} = \frac{1}{\pi} [\sin \alpha + (\pi - \alpha) \cos \alpha],$$

peaking at secondary eclipse and vanishing at transit (no nightside emission at all), with the dayside temperature corresponding to $f' = 2/3$.

Infrared phase curves: the general case

Beyond the simplest case, if the thermal flux distribution over the planet's surface is available from a model atmosphere or a circulation model, the thermal phase curve is still computed from the planet-star flux ratio, with the phase-dependent planetary flux

$$F_{\alpha,\text{thermal}}(\nu) = \int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} I_{\alpha,\text{thermal}}(\theta, \phi, \nu) \cos \phi \cos^2 \theta \, d\theta \, d\phi,$$

integrated over the hemisphere facing the observer at phase α (the subscript “thermal” denotes the thermal-infrared component of the intensity), and the flux ratio follows with R_p^2/R_*^2 and the stellar flux at that frequency.

- ▶ The shape of $I_{\alpha,\text{thermal}}(\theta, \phi)$ encodes the longitudinal temperature map: a hot spot shifted east of the substellar point produces a phase-curve peak *before* secondary eclipse (Lecture 1).
- ▶ The nightside flux sets the minimum of the phase curve near transit; the day–night amplitude measures how efficiently heat is redistributed, i.e. the effective f' .
- ▶ Real hot Jupiters (HD 189733b, HD 209458b) show amplitudes reduced relative to the no-redistribution Lambert curve and offset peaks: heat is carried to the nightside by winds. Ultra-hot Jupiters and planets on eccentric orbits (HD 80606b) show the full range from strong to weak redistribution.

The boundary conditions: a two-point problem

- ▶ To further investigate the hidden complication, return to the boundary conditions. To solve for the intensity from this *first-order* ordinary differential equation we require two full boundary conditions, that is, an upper and a lower boundary condition on the *full* range $-1 \leq \mu \leq 1$.
- ▶ Yet the information we have is the stellar radiation incident on the planet and traveling *downward*, $-1 \leq \mu < 0$, at the top, and an estimate of the interior energy that we may convert to an *outward-going* intensity, $0 < \mu \leq 1$, at the bottom.
- ▶ Because the boundary conditions are not fully specified — half of the rays are known at each boundary — iterative techniques or a different formulation of the transfer equation are needed to solve for I :
 - ▶ Λ iteration: guess S , compute I by the formal solution, update J and S , repeat (slow when $\omega_0 \rightarrow 1$; accelerated versions exist);
 - ▶ Feautrier's method: combine $I(\mu)$ and $I(-\mu)$ into symmetric and antisymmetric parts, giving a second-order equation with one condition at each boundary;
 - ▶ two-stream and discrete-ordinate methods (Heng Chapter 3), moment methods with a closure (Eddington approximation, Chapter 6), and Monte Carlo simulations.
- ▶ The same structure appears in every transport problem with scattering: the solution at a point depends on the solution everywhere else.

Beer's law

The simplest solution discards the source function ($S = 0$): pure extinction of an external beam.

$$\int_{I_0}^I \frac{dI}{I} = \int_0^\tau \frac{d\tau'}{\mu} \implies \boxed{I = I_0 e^{-\tau/|\mu|}}$$

(Heng writes $I_0 e^{\tau/\mu}$ with his $\mu < 0$ for penetrating rays; the intensity decreases either way.) This is **Beer's law** (or the Beer–Lambert–Bouguer law): the intensity is diminished exponentially, $\tau = 1$ corresponding to one e-folding, i.e. 63% of the beam removed ($1/e \simeq 37\%$ transmitted) when unit optical depth is traversed.

- ▶ The slant factor $1/|\mu|$: a beam entering at 60° from the normal traverses twice the optical depth of a normal beam. This is why the Sun is redder at sunset and why the transit chord is so opaque.
- ▶ Beer's law describes very well how *starlight* is absorbed on entering an exoplanet atmosphere, and it underlies transmission spectroscopy: the in-transit flux is the stellar flux times $e^{-\tau_{\text{chord}}(\nu)}$ integrated over the annulus.
- ▶ It is inadequate for the planet's *thermal emission*, which has both incoming and outgoing components and a source function that cannot be neglected.
- ▶ Heng Chapter 4 generalizes Beer's law to include non-isotropic scattering of the stellar beam.

Direct solution with pure absorption: a pair of layers

Beyond Beer's law, the equation can be solved directly for I only in the limit of *pure absorption*. Include coherent isotropic scattering, $S = (1 - \omega_0)B + \omega_0 J$, and integrate between two points 1 and 2 ($\tau_2 > \tau_1$):

$$I_2 e^{-\tau_2/\mu} - I_1 e^{-\tau_1/\mu} = -\frac{1}{\mu} \int_{\tau_1}^{\tau_2} [\omega_0 J + (1 - \omega_0)B] e^{-\tau/\mu} d\tau.$$

One cannot proceed unless $\omega_0 = 0$, because $J(\tau)$ is not known a priori. (“Coherent” means the wavelength is unchanged; spectral lines are better described by complete redistribution, but for bins wide compared to line widths coherent scattering is a decent approximation.)

Isothermal absorbing layer. Take $\omega_0 = 0$ and B, I_1, I_2 constant; label upward and downward radiation by $\uparrow, \downarrow$; integrate over the hemisphere and write fluxes as $F = \pi I$. Then

$$F_{\uparrow 1} = F_{\uparrow 2} \mathcal{T} + \pi B(1 - \mathcal{T}), \quad F_{\downarrow 2} = F_{\downarrow 1} \mathcal{T} + \pi B(1 - \mathcal{T})$$

with the **transmission function**

$$\mathcal{T} \equiv 2 \int_0^1 \mu e^{-\Delta\tau/\mu} d\mu = 2E_3(\Delta\tau) = (1 - \Delta\tau)e^{-\Delta\tau} + \Delta\tau^2 E_1(\Delta\tau), \quad \Delta\tau \equiv \tau_2 - \tau_1,$$

where $E_n(x) = \int_1^\infty t^{-n} e^{-xt} dt$ are the exponential integrals (Heng, problem 2.8.5).

The transmission function

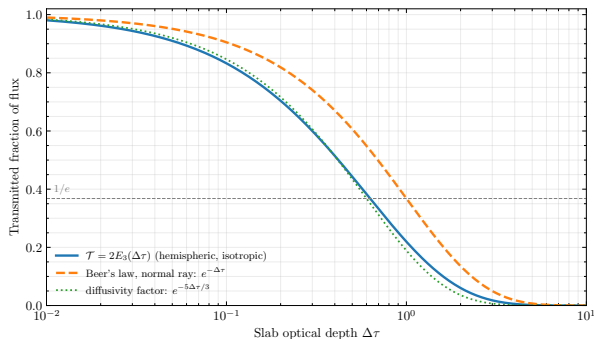


Figure 5: Hemispheric transmission $\mathcal{T} = 2E_3(\Delta\tau)$ of an isothermal slab compared with Beer's law for a normal ray and with the diffusivity-factor approximation $e^{-5\Delta\tau/3}$.

- $\mathcal{T}$ has a convenient range between 0 and 1. When $\mathcal{T} = 1$ no radiation is absorbed and no blackbody emission is produced: the incident flux passes right through the pair of layers, a completely transparent medium. When $\mathcal{T} = 0$ nothing passes through and the pair of layers radiates purely as a blackbody, $F = \pi B$.
- Because the flux is an average over all angles, and slant rays see a larger optical depth $\Delta\tau/\mu$, $\mathcal{T}$ falls faster than the normal-ray Beer's law $e^{-\Delta\tau}$: the slab reaches $1/e$ transmission at $\Delta\tau \simeq 0.6$, not at 1.

The transmission function: approximations and use

- ▶ **Diffusivity factor.** The classic approximation replaces the angle integral by a single ray at an effective angle, $\mathcal{T} \simeq e^{-D\Delta\tau}$ with $D = 5/3$ ($\mu_{\text{eff}} = 3/5$; Elsasser 1942). As the figure shows it is accurate to a few percent for all $\Delta\tau$, which is why single-angle (two-stream) methods work so well for thermal radiation.
- ▶ **Small $\Delta\tau$:** $\mathcal{T} \simeq 1 - 2\Delta\tau$ (not $1 - \Delta\tau$): a thin slab absorbs twice as much hemispheric flux as a normal ray would suggest, because the mean slant factor $\langle 1/\mu \rangle$ over the flux-weighted hemisphere is 2.
- ▶ **Large $\Delta\tau$:** $\mathcal{T} \simeq e^{-\Delta\tau}(1 - \Delta\tau) + \Delta\tau^2 E_1 \rightarrow 2e^{-\Delta\tau}/\Delta\tau$ asymptotically, dominated by the normal rays, the only ones that get through.
- ▶ **Pairs of layers.** Writing the solutions in this form describes the transfer of radiation from point 1 to point 2 and vice versa, using either location as the starting point or boundary condition. Formulating the problem as radiative transfer between *pairs of layers* is the basis of the two-stream treatment (Heng Chapter 3), and stacking many such layers is how a numerical longwave code computes the emergent flux for an arbitrary $T(\tau)$ profile.

Non-isothermal layers: a linear source function

The Planck function generally varies with optical depth. A direct solution still exists if B is expanded to first order about the layer boundaries (Heng, problem 2.8.5b):

$$B = \begin{cases} B_1 + B'(\tau - \tau_1), & \downarrow \text{ direction,} \\ B_2 + B'(\tau - \tau_2), & \uparrow \text{ direction,} \end{cases} \quad B' \equiv \frac{\partial B}{\partial \tau} = \frac{B_2 - B_1}{\tau_2 - \tau_1},$$

which reproduces B_1 at τ_1 and B_2 at τ_2 . The hemispherically integrated solutions become

$$F_{\uparrow 1} = F_{\uparrow 2} \mathcal{J} + \pi B_2 (1 - \mathcal{J}) + \pi B' \left[\frac{2}{3} (1 - e^{-\Delta\tau}) - \Delta\tau \left(1 - \frac{\mathcal{J}}{3} \right) \right],$$

$$F_{\downarrow 2} = F_{\downarrow 1} \mathcal{J} + \pi B_1 (1 - \mathcal{J}) - \pi B' \left[\frac{2}{3} (1 - e^{-\Delta\tau}) - \Delta\tau \left(1 - \frac{\mathcal{J}}{3} \right) \right].$$

- The extra term is the correction from the temperature gradient within the layer; it vanishes for $B' = 0$ and for $\Delta\tau \rightarrow 0$.

Eddington–Barbier and the formation of bands

- For a source function linear in optical depth, $S = a + b\tau$, the emergent-intensity integral gives *exactly*

$$I(0, \mu) = \frac{1}{\mu} \int_0^\infty (a + b\tau) e^{-\tau/\mu} d\tau = a + b\mu = S(\tau = \mu),$$

the **Eddington–Barbier relation**: the emergent intensity equals the source function at optical depth μ along the ray. For a general S it remains a good approximation.

- **Limb darkening.** Looking at the disk center ($\mu = 1$) we see $S(\tau = 1)$; near the limb ($\mu \rightarrow 0$) we see $S(\tau \rightarrow 0)$. If T decreases outward, the limb is darker; a stellar limb darkening curve therefore measures $dS/d\tau$, and modeling it is essential for transit radii (Lecture 1).
- **Absorption bands.** In a strong molecular band κ_ν is large, $\tau_\nu = \mu$ is reached high up where T is low, and $I(0, \mu, \nu) = B(T_{\text{high}})$ is small: the band appears in *absorption*. In a spectral window we see deeper, warmer layers and more flux.
- **Emission bands.** If T *increases* with height (a stratospheric inversion, as for ozone on Earth or TiO/VO on some hot Jupiters), the same bands appear in *emission*. The sign of the features in an emission spectrum thus diagnoses the sign of the temperature gradient.
- This is the whole basis of remote sensing of temperature structure: each frequency samples S , hence T , at its own $\tau_\nu \sim \mu$, i.e. its own pressure (Chapter 6 and Chapter 9).

Why the linear source function matters

- ▶ The “linear-in- τ ” source function is the standard building block of numerical longwave radiative transfer, for example in the widely used two-stream scheme of Toon et al. (1989) and in most exoplanet forward models: within each layer B is interpolated linearly in τ , and the layer solutions are chained together from the bottom boundary to the top.
- ▶ Integrated over a semi-infinite atmosphere, it gives the **Eddington–Barbier relation for the flux**: for $S = a + b\tau$, $F(0) = 2\pi \int_0^1 \mu(a + b\mu) d\mu = \pi(a + \frac{2}{3}b) = \pi S(\tau = \frac{2}{3})$. The emergent flux equals π times the source function at optical depth $2/3$ — the origin of the rule of thumb that the photosphere sits at $\tau \simeq 2/3$, and of the appearance of $2/3$ in grey-atmosphere temperature profiles.
- ▶ The same relation for the intensity gives $I(0, \mu) = S(\tau = \mu)$ (previous section), so the disk-center intensity samples $\tau = 1$ and the limb samples $\tau \rightarrow 0$: limb darkening measures $dS/d\tau$, i.e. the temperature gradient.
- ▶ With many thin layers, each isothermal or linear, the emergent flux of an arbitrary $T(\tau)$ profile can be built up layer by layer — exactly the forward model used in atmospheric retrievals (below), where speed matters because it is evaluated millions of times.

A practical checklist for radiative transfer calculations

What is needed to actually compute a spectrum (Heng, Section 2.5):

1. **Choose the form of the transfer equation** and the approximations (plane-parallel? scattering? two-stream or multi-stream?), check that the assumptions are plausible, and select the numerical method. Solutions are called *forward models*: they take abundances and opacities and produce a synthetic spectrum.
2. **Set up a model atmosphere** with a finite number of layers, each with its own T , P , and column mass, and compute the fluxes passing through each layer. The isothermal-layer solution above is a forward model in the pure-absorption limit; its inputs are the opacities, from which the optical depths, transmission functions, and fluxes of each layer follow.
3. **Decide the chemical composition** (Heng Chapter 7) and obtain the *opacities* of the atoms, molecules, and aerosols, typically from precomputed tables (Heng Chapter 5). Combining them, weighted by abundance, over T , P , and λ gives the opacity function of the atmosphere.
4. **Iterate to convergence**: opacities, optical depths, and temperatures are interdependent. Convergence is reached when the temperature of each layer stops changing (to within a numerical threshold).

The converged solution yields the flux at the top of the atmosphere (the synthetic spectrum) and the temperatures of all layers (the temperature–pressure profile).

Radiative equilibrium: local energy conservation

How are the temperatures and fluxes of the layers related? In the absence of atmospheric dynamics one hopes to solve for **radiative equilibrium**, which occurs when the net heat entering and leaving each layer vanishes. It is a statement of *local* energy conservation and originates from the first law of thermodynamics (Heng Chapter 9),

$$\rho c_P \frac{DT}{Dt} = \frac{DP}{Dt} + \rho Q,$$

with ρ the mass density, c_P the specific heat at constant pressure, and Q the heating term. Focusing on radiative heating (ignoring convection and conduction),

$$\rho Q = -\nabla \cdot \mathbf{F}_-,$$

where $\mathbf{F}_-$ is the wavelength-integrated *net* flux: the difference between the flux entering and exiting a layer, hence the minus subscript.

- ▶ Geometrically this makes sense: when radiation emanates from the layer, the divergence of the net flux is positive, which leads to cooling ($Q < 0$). The opposite happens when radiation enters the layer.
- ▶ The minus sign is crucial: if it is missing, one gets the unphysical situation of runaway cooling or heating.

Radiative equilibrium: the working equation

Ignoring work done on the system ($DP/Dt = 0$) and atmospheric dynamics ($D/Dt = \partial/\partial t$) gives the equation typically used to solve for radiative equilibrium in a one-dimensional model,

$$\boxed{\frac{\partial T}{\partial t} = -\frac{1}{\rho c_p} \frac{\partial F_-}{\partial z}} \quad (z \text{ increasing with altitude}).$$

- ▶ The computational iteration between the fluxes and temperatures, which depend on each other, is performed until the temperature stops changing in time (in practice until $\partial T/\partial t$ falls below a threshold, e.g. 10^{-6} ; an exact zero is impossible numerically). Then radiative equilibrium is formally obtained: $dF_-/dz = 0$, a constant net flux through every layer.
- ▶ In frequency-resolved form the same condition reads $\int_0^\infty \kappa_\nu (J_\nu - B_\nu) d\nu = 0$ in each layer: the energy absorbed equals the energy emitted. This is what fixes $T(\tau)$ and therefore B in the source function.
- ▶ Strictly speaking, if a calculation does not enforce radiative equilibrium, then we should not consider it a bona fide radiative transfer calculation of an atmosphere.

Local versus global energy conservation

- ▶ **Radiative equilibrium** is a condition on *every layer*: the net radiative flux through the atmosphere is the same at all heights, so no layer is heating or cooling. It determines the temperature–pressure profile.
- ▶ **Global energy conservation** is a condition on the atmosphere *as a whole*: the energy entering (absorbed starlight plus interior heat) equals the energy exiting (thermal emission to space). This is the energy balance of Lecture 2, which determines T_{eff} but not the profile.
- ▶ The two are often confused. Global conservation is a *necessary but insufficient* condition for radiative equilibrium: an atmosphere can have the right total emission yet an internally inconsistent profile in which some layers heat and others cool. A retrieval that enforces only the global condition may return such a profile.
- ▶ In the presence of **dynamics** radiative equilibrium is replaced by radiative–dynamical equilibrium: the divergence of the radiative flux is balanced by advection and by convection (where the radiative gradient exceeds the adiabatic one, Heng Chapter 4 and Seager Chapter 9). Even then the global condition holds for the planet as a whole, averaged over time.
- ▶ Heng Chapter 4 casts the distinction in precise mathematical terms using the double-grey solutions.

Clouds: the outstanding puzzle

The formation of aerosols or condensates in a gas is an outstanding puzzle of astrophysics (brown dwarfs, protoplanetary disks, exoplanets) and of Earth's climate science. Heng uses cloud, haze, aerosol, and condensate interchangeably, aware of the subtle differences: hazes are usually photochemical products; clouds follow the thermodynamic condensation curves. Forming clouds from first principles requires:

- ▶ **Nucleation:** how the precursor seed particles form, on which further growth proceeds; a true theory predicts the shape, composition, and size distribution of the particles.
- ▶ **Self-consistent gas and solid chemistry:** as the gas is depleted to form solids, its atomic and molecular abundances adjust.
- ▶ **Self-consistent opacities and radiative transfer** for gas and cloud particles, with the absorption and scattering of each computed from first principles across λ , T , and P rather than by ad hoc parameters.
- ▶ **Dynamics:** to stay aloft, the terminal velocity of a particle under gravity must be balanced by upwelling flow; sedimentation removes condensates below the photosphere. Usually parametrized by an eddy mixing coefficient K_{zz} .

Forming the particles is hard; once formed, modeling their effect on the spectrum is not controversial. “Aerosol” is the neutral term: any particle described by its size, composition, and index of refraction (or extinction efficiency).

What clouds do to a spectrum

To lowest order the aerosol adds an extinction efficiency $Q_e(x)$ with $x = 2\pi r/\lambda$ (the fitting formula and Heng Fig. 2.2 above):

- ▶ For $x \gg 1$, Q_e is roughly constant: a *grey* continuum opacity. For $x \ll 1$ it is Rayleigh-like, $\propto x^4$.
- ▶ The parameter Q_0 is a proxy for composition: refractory species (silicates) have $Q_0 \approx 10$, volatile ones (water) $Q_0 \approx 40$ –80. Particle radius has a somewhat larger influence than composition.
- ▶ A cloud particle influences absorption and scattering only at $\lambda \lesssim 2\pi r$. This generalizes to a size distribution.

Consequences for spectra. When the cloud optical depth approaches unity above the gas photosphere, spectral features and lines become *muted* and less sharp for $\lambda \lesssim 2\pi r$ — the flat transmission spectra of GJ 1214b and the weakened water bands of many hot Jupiters. But the same muting can be produced by *lower* abundances of the absorbing species, so clouds introduce a **degeneracy** in the interpretation of chemical abundances. To the next order, particles are described by ω_0 and g_0 , layer and wavelength dependent, in a two-stream or multi-stream calculation.

Atmospheric retrieval: letting the data talk

Traditionally, when one builds a model of an atmosphere, one starts with a governing equation — of fluid dynamics, radiation, chemistry, or some combination — and a set of assumptions, and computes. Ideally the calculation predicts a measurable quantity that may be confronted by the observations, and a judgment is made on whether the model has represented Nature — but only against the preconceived notions we have about an atmosphere.

- ▶ **Atmospheric retrieval** tries to shed those notions by providing a framework in which one *inverts* a set of observables (a spectrum) to obtain the properties of the atmosphere: its thermal structure and chemical abundances.
- ▶ Origin: inversion techniques for remote-sensing observations of Earth (Rodgers 2000), later used in the planetary sciences for the Solar System bodies. There, a combination of satellite and in-situ data provides ground truth to anchor the retrieval. “We can simply let the data do the talking.”
- ▶ The output is not a single model but a *posterior distribution*: the range of atmospheres consistent with the data and their uncertainties.

Why retrieval is harder for exoplanets

- ▶ For exoplanets the faith in the data alone is less viable: we are dealing with remote sensing of *unresolved point sources*, and in-situ measurements are out of the question.
- ▶ The data are often sparse and fragmented, in the sense that one only sees incomplete facets of the atmosphere — the dayside but not the nightside, insufficient wavelength coverage, a handful of photometric bands — that do not allow unique interpretations.
- ▶ Letting the interpretation be guided entirely by the data, for example without enforcing radiative or chemical equilibrium, becomes foolhardy when the number of model parameters exceeds the number of data points. Our interpretation needs to be anchored by the laws of physics and chemistry to some degree.
- ▶ How much physics to impose is an ongoing debate: “free” retrievals with parametrized profiles and abundances versus “self-consistent” models with equilibrium chemistry and radiative equilibrium. The former are flexible but can wander into unphysical corners; the latter are constrained but may be wrong if the assumed physics is incomplete.
- ▶ JWST-quality spectra, with hundreds of spectral points, have shifted the balance: the data now have enough information to test the assumptions of the self-consistent models rather than merely to fit a few parameters.

Basic ingredients of a retrieval model

Augmenting the radiative-transfer checklist:

1. A way of describing the **temperature–pressure profile** with a set of parameters: either an ad hoc fitting function, or a simplified analytic solution of the radiative transfer equation (Heng Chapter 4, the double-grey solutions).
2. The **identities of the atoms and molecules** included (e.g. water, methane, carbon monoxide). Their relative abundances may be free parameters of the retrieval; alternatively one may enforce *chemical equilibrium*, in which case the molecular abundances are determined by the elemental abundances and only the latter are free (Heng Chapter 7).
3. For a fixed combination of parameter values, a **forward model** converts the opacities into a synthetic spectrum. This conversion is performed once for each parameter set; there is typically no attempt to solve for radiative equilibrium, as the iteration described earlier is not performed.

Scanning the parameter space

- ▶ Millions, if not billions, of combinations of these parameters are considered. Each synthetic spectrum is compared against the observed one and a goodness of fit (a likelihood, typically $\propto e^{-\chi^2/2}$) is computed for each model.
- ▶ In this manner a range of best-fit models is found, given the uncertainties on the data, and the *posterior distributions* of the molecular abundances and of the temperature profile may be computed. Correlations between parameters (the degeneracies of the next slides) show up as elongated or banana-shaped posteriors.
- ▶ In practice a minimization or sampling algorithm is required to scan this large, multidimensional parameter space efficiently and thoroughly:
 - ▶ **Optimal estimation** exploits the property that if the prior of an input quantity is Gaussian, the expressions used for minimization simplify (Rodgers 2000). A uniform prior is mimicked by a very wide Gaussian.
 - ▶ **Monte Carlo methods** (Markov-chain Monte Carlo, nested sampling) allow general priors and have been implemented in various flavors; they are now standard in exoplanet retrievals.
- ▶ Because the forward model is evaluated millions of times, its speed is critical: this is why retrievals use fast layer-by-layer pure-absorption solutions with precomputed opacities rather than full radiative-equilibrium iterations.

Retrieval pitfalls (1)

A retrieval model is only as good as the assumptions and techniques used. Some of the possible pitfalls:

- ▶ **Degeneracies.** The spectral lines of different molecules inevitably overlap across wavelength, so multiple combinations of relative abundances may lead to the same net absorption at a given wavelength. Aerosols introduce another degeneracy: they diminish the strength of spectral lines, an effect that may be compensated by increasing the abundances of the molecules responsible. Any good retrieval quantifies these degeneracies formally and systematically (correlated posteriors).
- ▶ **Being misled by your priors.** The posteriors depend on the priors assumed. Example: the carbon-to-oxygen ratio estimated as

$$C/O \approx \frac{\tilde{n}_{CO} + \tilde{n}_{CH_4} + \tilde{n}_{CO_2}}{\tilde{n}_{CO} + \tilde{n}_{H_2O} + 2\tilde{n}_{CO_2}},$$

with $\tilde{n}$ the mixing ratios (abundances normalized by H_2). If CO dominates, $C/O \approx 1$; if CO_2 dominates, $C/O \approx 0.5$; if CH_4 or H_2O dominate, $C/O \gg 1$ or $\ll 1$. So even uniform priors on the molecules produce two artificial peaks in the prior distribution of C/O .

Retrieval pitfalls (2)

- ▶ **Local versus global energy conservation.** In the ensemble of models it is natural to discard those whose emergent thermal emission exceeds the energy input from the star (for negligible internal heat). That is simply global energy conservation: you cannot produce more than what you put in. What is less obvious is *local* energy conservation, radiative equilibrium: the net heating or cooling of each layer must be the same across layers, in the absence of circulation, and most retrievals do not enforce it.
- ▶ **Not everything is chemically possible.** In chemical equilibrium not every combination of relative abundances is possible: each layer has its own T and P , and the mixing ratios take specific values given the elemental abundances. One is not free to specify them as independent parameters. When dynamical mixing is faster than chemistry the mixing ratios depart from local equilibrium — but again not arbitrarily: with the *quenching* approximation (Heng Chapter 6) they originate from a deeper layer that *is* in equilibrium. Example: CO_2 in hot ($\gtrsim 1000$ K) hydrogen-dominated atmospheres is never more abundant than CO or H_2O unless the metallicity is implausibly high.

Retrieval pitfalls (3)

- ▶ **Opacities: the devil is in the details.** Across the UV to infrared one must compute the shapes of millions to billions of lines, for one molecule at one (T, P) . Line-by-line calculations sample all of them without approximation, a formidable task at ~ 1000 K; binning techniques such as the k -distribution method speed this up at the cost of approximations (Heng Chapter 5). Hidden in the details is the pressure broadening of the far line wings, an unsolved physics problem. And photometric observations integrate over broad bandpasses, so many combinations of molecules can produce the same opacity across the band. When published papers omit these details, exercise skepticism — one of the most precious instruments in a scientist's toolkit.
- ▶ **Numerical convergence.** Running a retrieval requires discretizing space (the T - P profile), wavelength (the opacity function), and the set of species; to finish in reasonable time one often uses coarse resolution along one axis. Determining whether the resolution is sufficient requires resolution tests — thankless but necessary. As a rule of thumb (attributed by Heng to Scott Tremaine), always adopt the second-highest resolution if the highest one attains convergence.

Take-home messages

Albedos. A perfect Lambert surface has $p = 1$ by energy conservation; a Lambert sphere has $A_g = 2/3$, $q = 3/2$, $A_S = 1$. A_g can exceed 1; A_B is A_S weighted by the stellar spectrum and enters the energy balance.

Flux ratios. Thermal: $F_p/F_* = (R_p/R_*)^2 B(T_p)/B(T_*)$, $\rightarrow (R_p/R_*)^2 T_p/T_*$ in the Rayleigh–Jeans limit. Scattered light at full phase: $F_p/F_* = A_g(R_p/a)^2$. Transmission: the annulus of a few scale heights, $\sim 2R_p H/R_*^2$.

Phase curves. $\Phi_\alpha = \Psi_\alpha/\Psi_{\alpha=0}$; a Lambert sphere has $\Phi_\alpha = [\sin \alpha + (\pi - \alpha) \cos \alpha]/\pi$. Infrared phase curves map the day–night temperature contrast.

Simple solutions. Beer’s law for pure extinction; isothermal layers exchange flux through $\mathcal{T} = 2E_3(\Delta\tau)$; a linear source function gives Eddington–Barbier. Radiative equilibrium is *local* energy conservation.

Clouds and retrieval. Clouds add grey extinction and degeneracies; retrieval inverts spectra but must respect physics and chemistry.

Next. Line formation, scattering atmospheres, and the two-stream approximation (Lecture 6).

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